INTERNSHIP PROJECT REPORT

Real-World Dataset Analysis Project

Analysis of the Global Superstore Sales Dataset

Python | Jupyter Notebook | Pandas | NumPy | SciPy | Matplotlib | Seaborn

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Course / Year:	**B.Tech, 3rd Year**
College / University:	**Invertis University**
Internship Provider:	**YuvaIntern**
Internship Period:	**October 01, 2026 to November 26, 2026**
Date of Submission:	**[SUBMISSION DATE]**

REPORT TEMPLATE - RESULTS PENDING EXECUTION. The structure, methodology and explanations are final. Every place marked [PENDING EXECUTION] is filled automatically with the real numbers, tables and charts when you run the notebook (or python src/analysis.py) and then python src/build_report.py.

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1. Project Title and Summary

Real-World Dataset Analysis Project is an internship project completed through YuvaIntern. It analyses a real-world retail dataset, the Global Superstore Sales dataset, to find where sales and profit come from, where money is lost, how sales change over time and how discounts affect profitability. The work covers dataset selection, environment setup, data exploration, cleaning, exploratory data analysis (EDA), statistical analysis, visualization and business recommendations, using Python, Pandas, NumPy, SciPy, Matplotlib and Seaborn in a Jupyter Notebook.

Summary of results: [PENDING EXECUTION]

2. Objectives and Problem Statement

Problem statement

Using the order data of a global retailer, which categories, sub-categories, products, markets and customers generate the most sales and profit, how do sales change over time, and how do discounts affect profitability? The answers should lead to practical, evidence-based recommendations.

Objectives

1. Select a reliable real-world dataset and understand its business context.
2. Explore the dataset structure, data types, distributions, missing values, duplicates and relationships.
3. Clean and preprocess the data appropriately and document every change.
4. Perform exploratory data analysis (EDA) using Pandas, NumPy and statistical techniques.
5. Create meaningful visualizations using Matplotlib and Seaborn.
6. Present findings, interpretations, conclusions and recommendations in a professional notebook and report.

3. Dataset Source and Description

Item	Details
Dataset	Global Superstore (file Global_Superstore2.csv; Excel version in the same download)
Source	Kaggle, dataset apoorvaappz/global-super-store-dataset (https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset)
Documented size	51,290 rows x 24 columns (as reported by published notebooks that use this file)
Size in this analysis	[PENDING EXECUTION]
License	Not verified from outside Kaggle; check the License field on the Kaggle page. The data file is therefore not redistributed in the project repository.
Business context	Order lines of a global retailer selling products in several categories to customers in many countries and markets.

Important columns: Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Segment, Country, Market, Region, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit, Shipping Cost and Order Priority. Column meanings follow the public dataset descriptions. The dataset is suitable

because it contains dates, categories, numerical measures and several business dimensions (product, market, customer, shipping), and has enough records for meaningful analysis.

4. Methodology

- **Environment:** Python, a virtual environment, Jupyter Notebook, Pandas, NumPy, SciPy, Matplotlib and Seaborn (requirements.txt).
- **Loading:** the file is read with automatic text-encoding detection; column names are standardised; a schema check reports missing columns.
- **Exploration:** head/tail, shape, data types, descriptive statistics, missing values and percentages, duplicates, unique values, category counts, skewness and date inspection.
- **Quality assessment:** missing values, duplicates, impossible values (negative sales, zero quantity, discount outside 0-1, shipping before ordering) and outliers by the IQR rule.
- **Cleaning:** seven documented steps on a copy of the data; the original is never modified; an audit log records the rows affected by each step.
- **EDA:** KPIs, sales and profit by category, sub-category, market, region, time, product, customer and segment, discount analysis and loss-making items.
- **Statistics:** descriptive statistics, skewness and kurtosis, Pearson and Spearman correlation, Kruskal-Wallis and Mann-Whitney tests (non-parametric, because the data is skewed).
- **Visualization:** 13 charts with titles, labelled axes and interpretations.
- **Reproducibility:** one command (python src/analysis.py) repeats the whole analysis; automated checks are in tests/test_pipeline.py.

5. Data Cleaning Procedures

Step	Description	Rows before	Rows after	Rows removed	Details
1	Strip spaces from text columns	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]
2	Convert data types (dates, numbers); check discount scale	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]
3	Handle missing values	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]
4	Remove exact duplicate rows	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]
5	Remove impossible values	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]
6	Create new columns	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]
7	Flag outliers (IQR rule) - kept in data	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]	[PENDING EXECUTION]

Decisions (design of the cleaning, final): exact duplicates and impossible records are removed; small amounts of missing data are filled (median for numbers, 'Unknown' for text); columns with 5% or more missing values are kept but not used; outliers are flagged and kept because they are genuine business events. The original data is preserved.

6. Exploratory Data Analysis Results

Key figure	Value
Period	[PENDING EXECUTION]

Key figure	Value
Orders / order lines	[PENDING EXECUTION]
Customers	[PENDING EXECUTION]
Total sales	[PENDING EXECUTION]
Total profit	[PENDING EXECUTION]
Profit margin	[PENDING EXECUTION]
Average / median order value	[PENDING EXECUTION]
Average discount	[PENDING EXECUTION]
Loss-making order lines	[PENDING EXECUTION]

Tables of sales and profit by category, market, year and discount band, and the results of the statistical tests (Spearman, Kruskal-Wallis, Mann-Whitney): [PENDING EXECUTION]

7. Charts and Interpretations

Thirteen charts are inserted here after execution ([PENDING EXECUTION]): sales and profit by category; profit by sub-category; sales and profit by market; profit-margin heatmap; distributions of sales and profit; monthly trend; yearly sales by category; discount versus profit; profit box plot; correlation heatmap; top and bottom products; shipping analysis; year-month sales heatmap. Each chart is followed by a written interpretation.

8. Key Findings and Evidence

Ten findings are generated from the results ([PENDING EXECUTION]). Each finding has four parts: **observation**, **numerical evidence**, **business significance** and **recommendation**. Topics: overall performance; category; sub-category and loss-makers; market/region; trend and seasonality; top and loss-making products; customers and segments; discount versus profit; shipping; outliers.

9. Business Recommendations

Recommendations derived from each finding: [PENDING EXECUTION]

10. Limitations

- **Association, not causation:** correlations and group differences describe the data; they do not prove cause and effect.
- **Single dataset:** the results describe this file and may not hold for other periods, companies or countries.
- **Unknown definitions:** the exact meaning of Profit and the currency are not documented in the sources that could be verified.
- **Outliers kept:** extreme orders influence totals and averages; medians and non-parametric tests were used to reduce the effect.
- **Unused columns:** columns with many missing values (for example postal code, if affected) were not used.
- **Descriptive only:** no forecasting or machine-learning model was built.
- **Drafted insights:** the finding text is generated by rules from the results and was reviewed by the student.

11. Conclusion

Conclusion based on the real results: [PENDING EXECUTION]

12. References

7. Kaggle dataset: Global Superstore, apoorvaappz/global-super-store-dataset. https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset
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